On-Demand Scheduling of Hybrid Vehicles For Parcel Delivery in a Smart City

Project report submitted for the partial fulfilment of the requirement for the degree of **Bachelor of Technology** in **Computer Science and Engineering** from the University of Calcutta.

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Certificate

This is to certify that **Prateek Ganguli**, **Vatsal Dadia**, and **Hridita Basu**, students of the 8th semester of Bachelor of Technology in Computer Science and Engineering in the Department of Computer Science and Engineering from the University of Calcutta, have completed their project report titled **"On-Demand Scheduling of Hybrid Vehicles For Parcel Delivery in a Smart City"** under my supervision and guidance, for the partial fulfilment requirement for the degree of Bachelor of Technology in Computer Science and Engineering from the University of Calcutta.

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Preamble

Problem Statement: For a given portion of the map for a geographic region, we have a central warehouse (assumed to have stocks of all parcels that customers could request). We also have a set of drone pickup points around which the customers' residences are clustered. Each pickup point is serviced by a single drone. The routes that volunteering third-party cabs will take with their passengers are known in advance. Online parcel requests come from customers for which third-party cabs are assigned to deliver parcels from the warehouse to a chosen pickup point. The parcels dropped off at a given pickup point are then delivered to the respective customer residences around the pickup point by the drone servicing that pickup point. Both cabs and drones have a limited capacity for carrying parcels. Drones additionally also have a limited flying range that they can cover in a single flight without returning to the pickup point for a battery swap. The objective is to assign a cab and a pickup point for each parcel, followed by scheduling a Travelling Salesman Problem (TSP) route for the drone at the chosen pickup point, such that the overall cost of delivery (measured by a fee charged as per the distance that the parcel was transported by cab and by drone) is lowered than if the parcels were delivered using only first-party owned delivery trucks.

Domain: This project lies within the domain of optimization and also sustainable parcel deliveries. It focuses on creating an efficient algorithm for the delivery of parcels using a hybrid fleet of third-party cabs and first-party drones, while operating within the constraints of the parcel-carrying capacities of cabs and drones, in addition to the limited flying range of drones on a fully charged battery.

Completed Work: We have studied several publications which propose methods of drone deliveries and identified their shortcomings. Inspired by data caching in content-centric Vehicular Ad-hoc networks (VANETs), we propose a novel paradigm of deliveries using a hybrid fleet of third-party cabs and first-party drones. We have obtained experimental results of our algorithm by simulating cab trips to the warehouse and to the pickup points, along with the drone routes centred around the pickup points. From these simulations, we report the total distances that each parcel was transported by a cab, followed by a drone, and the time step at which the delivery of the drone was completed at the corresponding residence from which the parcel was requested.

Acknowledgement

We are pleased to express our gratitude to all those who have extended their support and encouragement to help us carry out our project work. We express our reverence and gratitude to our project guide Dr. Sunirmal Khatua, Professor of the Department of Computer Science and Engineering, University of Calcutta, for his guidance, constructive criticism, and constant encouragement throughout the project. We would also like to thank Dr. Rajib Kumar Das, Professor of the Department of Computer Science and Engineering, University of Calcutta, for his kind assistance with all our research queries, and Prof. Preetam Kumar Sur for his guidance with the simulation tools used. We would also like to acknowledge the researchers and scholars who have significantly contributed to the field of operations research. Their pioneering work and published papers served as a valuable foundation for our project, and we are grateful for their dedication to advancing knowledge in this domain.

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Chapter 1

Introduction

In the setting of a smart city, we have several key technologies available to use, one of which comprises Internet-of-Things (IoT) enabled platforms such as Unmanned Aerial Vehicles (UAVs), which have several advantages: fast, flexible, lightweight, low-maintenance, and easy to deploy. As a result, UAVs have seen rapid adoption in applications such as various civilian and military fields, including sensing and inspection, disaster warning, wireless sensor networks, cargo delivery, and so on.

Such kinds of applications often require UAVs to work across a wide area and visit multiple different locations to execute data collection or cargo distribution. However, UAVs are limited by battery capacities, and maximum possible thrust produced, thus putting an upper bound on the maximum flying range and the amount of cargo they can carry. In order to circumvent these limitations, people customarily utilize ground vehicles to carry and assist UAVs in various applications, including cargo delivery. This vehicle-drone cooperative delivery mode benefits from both the large capacity of goods of ground vehicles and the high mobility of UAVs.

In recent years, e-commerce businesses have seen an increase in the daily volume of parcels to be delivered. In this respect, the delivery mechanism becomes prohibitively expensive, particularly for the final kilometre. To stay competitive and meet the increased demand, businesses have opted for innovative autonomous delivery options for the last mile, such as UAVs.

Vehicular Ad-Hoc Networks (VANETs) have extensively been studied in the context of cooperative caching and delivery of data packets in the case of content-centric networks. Inspired by this, we propose a novel paradigm of parcel deliveries using a hybrid fleet of third-party-owned cabs and first-part owned drones, where cabs 'cache' the parcels on their way from the central warehouse to one of the numerous drone pickup points. Subsequently, the drone servicing each pickup point traverses a Travelling Salesman route to deliver respective parcels from the pickup point to the surrounding residences.

Chapter 2

Background and Motivation

Literature	Problem Addressed	Proposed Strategy	Limitation
Grippa et al. (2018)	Drone assignment. Task assignment.	Job first to random vehicles. Job first to nearest vehicles.	Limited number of drones.
Sawadsitang et al. (2018)	Drone assignment. Supplier cooperation policy to share the total cost service.	Algorithm deciding where to assign batteries and chargers. Determine the time instance at which a battery begins or stops charging.	Transfer package among supplier depots increases the cost of delivery; if supplier does not have homogeneity of goods, the delivery model will be complex.
Peng et al. (2019)	Drone cargo delivery.	Genetic algorithm which combines solutions which use carriers to transport multiple drones to set location to carry out deliveries.	No real-world tests. The number of drones to be carried is not specified, which is a problem when the number of customers is unknown.
Das et al. (2021)	On-the-go parcel delivery using drones and trucks.	Integrating UAVs with delivery trucks, transforming the trucks into mobile launching and charging stations for UAVs.	The model is limited to deterministic demand and fixed truck and drone pairs.

Table 2.1: An overview of some relevant works from the literature on drone delivery systems.

To explore the problem space in our chosen direction of work, and to form an understanding of the background of existing research in this domain, we studied several published works of research [4, 1, 3, 2] concerning deliveries conducted using land vehicles and drones. A discussion of the studied materials, as summarized in Table 2.1, follows. Overall, the studied literature can be broadly classified into the categories depicted in Figure 2.1.

After careful study of these works of research and their corresponding shortcomings, we provide a discussion of a mathematical formulation of the problem

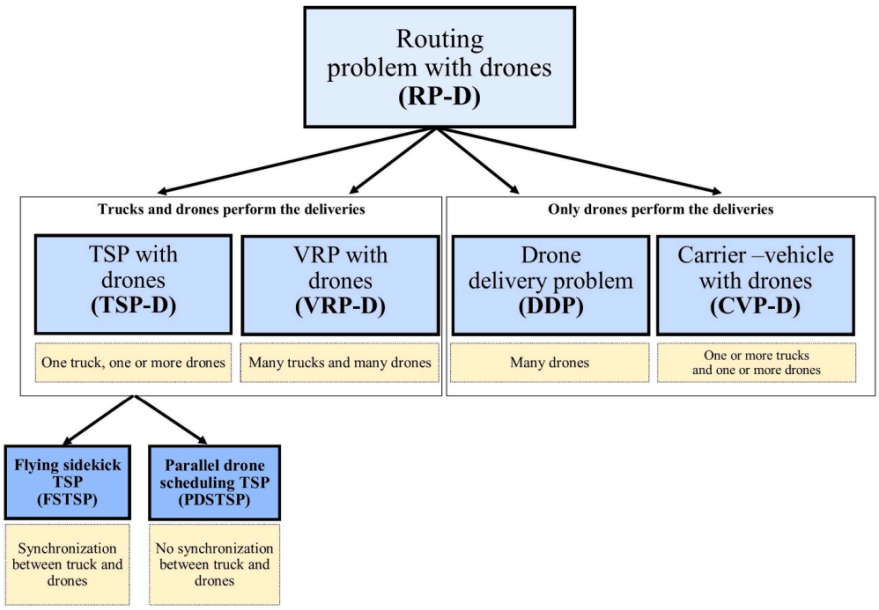

Figure 2.1: Classification of routing problems with drones.

statement. We also provide a description of the suggested algorithms for solving the proposed problem statement, along with experimental results obtained from simulations conducted using the specified research setup.

Grippa et al. [1] provides a formulation for delivery systems that use Unmanned Aerial Vehicles (UAVs) as a Vehicle Routing Problem (VRP), which is a generalization of the Travelling Salesman Problem (TSP). The vehicle routing problem (VRP) is a combinatorial optimization and integer programming problem which asks "What is the optimal set of routes for a fleet of vehicles to traverse to deliver to a given set of customers?"

Usually, a Vehicle Routing Problem has an instance which is formulated as having the following constraints:

1. all customers are supplied with the requested products
2. none of the vehicles exceeds its capacity traveling along its route
3. no customer is visited more than once
4. all routes start and end at one central depot
5. the overall routing cost is minimized.

In practical applications, VRPs have a broad diversity of additional requirements and operational constraints affecting the construction of the optimal set of routes. As such, their paper provides the following contributions, wherein they extend upon the basic constraints of VRP in the context of UAVs:

1. They give a lower bound for the expenditure needed to set up a stable system as a function of the targeted average delivery time.

2. They analyze two simple job assignment policies: The nearest job first to a random vehicle (NJR) and the first job first to the nearest vehicle (FJN). These policies are sub-optimal but their performance yields insight on the system dynamics which helps us to design advanced policies.

3. They analyze a policy known from queuing theory, named first job to the vehicle with the smallest workload ($FJW\pi$) in this paper, and modify it according to the lessons learned from NJR and FJN. We propose the first job to the vehicle with the smallest additional workload ($FJW\delta$) policy. $FJW\pi$ is not optimal in light load and stabilizes the system only for a fraction of the possible arrival rates. The novel $FJW\delta$ is optimal in light load and stabilizes the system for almost all arrival rates.

4. They show that FJN and FJW exhibit a tipping point behavior: One vehicle makes the difference between almost optimal performance and instability. Therefore, careful dimensioning is required. In contrast to this, $FJW\delta$ can stabilize the system for almost all arrival rates, as long as the number of vehicles per depot is sufficient. This finding yields a connection between dimensioning and control of the system.

5. Their last finding enables proper dimensioning of the delivery system controlled with $FJW\delta$.

If all system input is known before the vehicles leave the depot(s) and does not change during mission execution, the matter of concern is said to be both static and deterministic. For many real-world applications at least some input information like customer arrivals behave according to a probability distribution rather than is known a priori. These VRPs are denoted as stochastic. If some input information appears or changes during mission execution and has to be integrated immediately into decision-making, the VRP is called dynamic. Then, designing sets of routes has to be replaced by designing routing policies which describe the evolution of motion paths as a function of newly arriving input.

In their paper, the arrival of delivery requests for goods within a service area is generated by a Poisson process with an arrival rate of λ. The requests, also called jobs, are indexed by the job identifier n, which indicates the order of request arrivals. The corresponding customer is called c_n, and his or her position is denoted by xy coordinates and assumed to be independently and uniformly distributed in the service area. The information about the waiting customers is centrally stored

and is made available to all entities in the system. An example timeline of processes that a particular drone might have to go through is depicted in Figure 2.2. For each of the scheduling policies considered using the randomly distributed Poisson model of customer requests, several simulations were carried out. The data collected from the simulations were used to evaluate the average delivery time obtained for each such scheduling policy considered.

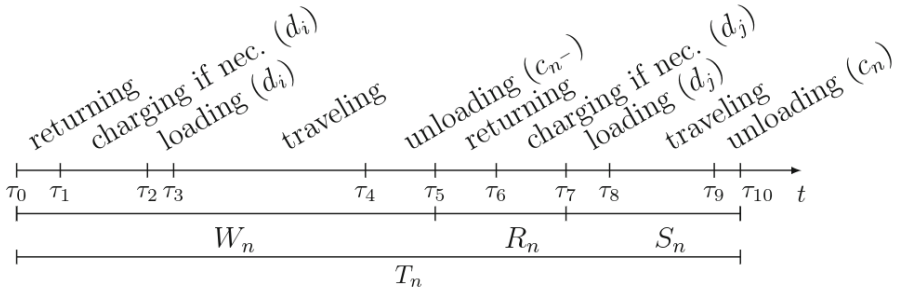

Figure 2.2: Time intervals involved in customer service: delivery time T, waiting time W, return time R, and service time S.

The paper by Sawadsitang et al. [2] discusses the increasing popularity of drones in various fields, emphasizing their potential in parcel delivery. While drones offer advantages such as faster delivery and lower costs compared to ground-based vehicles, they face limitations like flying distance and payload capacity. To address these challenges, the paper proposes a Supplier Cooperation in Drone Delivery (CoDD) framework. This framework involves three decision-making components: supplier cooperation, cost management, and parcel assignment. The merge-and-split algorithm is used for coalition structure among suppliers, the Shapley value for fair cost distribution, and a mixed-integer programming problem for optimizing the trade-off between drone and carrier delivery. The paper evaluates the CoDD framework using datasets, highlighting the influence of the number of customers on cooperation decisions. The objective is to assist rational suppliers in making optimal decisions to minimize costs and enhance stability in drone delivery.

The system model and assumptions are outlined for a scenario involving multiple suppliers using drones for parcel delivery. Suppliers have two options: using their own drones or outsourcing delivery to a carrier. Suppliers can cooperate, forming a coalition with shared customers and drones. The CoDD framework components, including Parcel Assignment, Supplier Cooperation, and Cost Management, are described in Figure 2.3.

In Parcel Assignment, the supplier has to decide on which drone is to deliver which parcel. Let $N = \{p1, p2, \ldots, p|N|\}$ be a set of all suppliers, where $|N|$ is the total number of suppliers. A supplier $p \in N$ has one depot. C_p and D_p denote

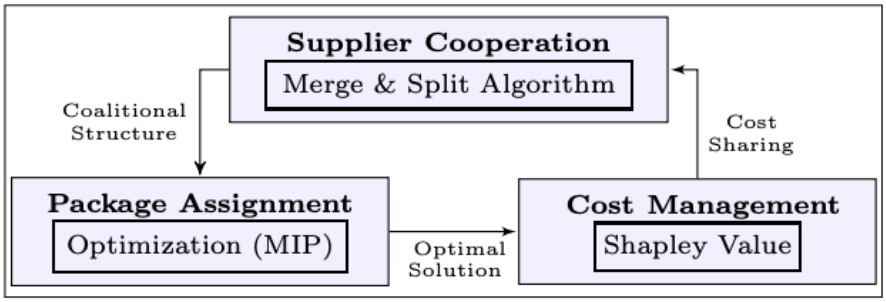

Figure 2.3: The supplier cooperation in CoDD framework.

the set of customers and the set of drones of supplier p, respectively. Two or more suppliers can cooperate and form a coalition. Let P be a set of suppliers in the same cooperation, i.e., coalition $P \subseteq N$. When suppliers cooperate, the suppliers will share their customers to serve and their drones which are referred to as a pool. A matrix O is used to represent which customer belongs to which supplier, and the element of the matrix denoted by $O_{i,p}$ is a binary parameter. $O_{i,p} = 1$ indicates that customer i belongs to supplier p, and $O_{i,p} = 0$ otherwise. Additionally, the parcel of customer i has a specific weight, which is denoted as a_i. Every drone d has its limits, i.e., daily flying distance limit l_d, flying distance limit per trip e_d, capacity limit f_d, and working hour limit h_d. Every drone d has an average flying speed, which is denoted as s_d. The flying distance from location i to location j is denoted as $k_{i,j}$. There are four costs involved in the parcel assignment: the initial cost of a drone (such as rental fee, preparation cost, and manpower), routing cost (which is incurred from travelling such as fuel or energy cost), the transferring cost which is incurred when one supplier ships or picks up parcels from other suppliers in the same pool, and outsourcing cost paid to a carrier.

In Supplier Cooperation, given the set of all suppliers N, a certain supplier chooses to cooperate with other suppliers by forming a coalition denoted as a set P. $\phi = \{P_1, P_2, \ldots, P_{|\phi|}\}$ denote a coalition structure, where $|\phi|$ is the total number of coalitions in the coalition structure. The coalition structure is a set of all coalitions that include all the suppliers, i.e., $N = P_1 \cup P_2 \cup \ldots \cup P_{|\phi|}$.

Given $N = \{p_1, p_2, p_3, p4\}$ and $\phi = \{\{p_1, p_2\}, \{p_3, p_4\}\}$, this coalition structure consists of two coalitions, i.e., $P_1 = \{p_1, p_2\}$ and $P_2 = \{p_3, p_4\}$, where the supplier p_1 cooperates with the supplier p_2 and the supplier p_3 cooperates with the supplier p_4. The total number of possible coalition structures can be calculated by the Bell number based on $|N|$.

The objective of the optimization problem of the parcel assignment, which is formulated as a mixed integer programming, is to minimize the total delivery cost. The optimization problem has six binary decision variables: W_d which indicates whether the initial cost of drone d needs to be paid or not, $Y_{i,d,p,q}$ is an allocation

binary variable of drone d where $Y_{i,d,p,q} = 1$ if drone d departs from the depot of supplier p to serve customer i and continues travelling to the depot of supplier q or $Y_{i,d,p,q} = 0$ otherwise, Z_i indicates whether customer i will be served by an outsourcing carrier or not, $M_{i,p,q}$ is a transferring binary variable where $M_{i,p,q} = 1$ if the parcel of customer i needs to be transferred from the depot of supplier p to the depot of supplier q, T_p indicates whether the transferring cost needs to be paid or not(the transferring cost is not paid by for an individual supplier but is included in the delivery cost of the pool) and $B_{p,d}$ is an auxiliary variable for eliminating impractical route where $B_{p,d} = 1$ if drone d departs from the depot of supplier p to serve a customer and comes back to the same depot.

To summarize, the supplier cooperation in drone delivery (CoDD) framework has been proposed to help suppliers decide whether to cooperate with other suppliers or not. The optimization is proposed to address the trade-off between drone delivery and outsourcing parcels to a carrier. The merge-and-split algorithm and Shapley value have been adopted to decide how suppliers should cooperate and share the cost incurred from the delivery and cooperation. Moreover, the framework has been evaluated by using data from the Solomon Benchmark suite and Singapore real data which shows that if the number of customers in the serving area is small, then outsourcing parcels to a carrier is cheaper than drone delivery and when the number of customers in the serving area is large, the initial cost of drones and the locations of depots have an impact to the stable coalition structure. The shortcomings mentioned in the paper include the flying distance limit per trip, small delivery capacity, inability to carry heavy parcels, and the higher cost of outsourcing parcel delivery to a carrier. These factors present challenges for drone delivery and need to be addressed by suppliers.

Peng et al. [3] investigate a routing problem for vehicle-drone cooperative parcel delivery. They aim to overcome the issues caused due to the limited battery capacities of Unmanned Aerial Vehicles (UAVs) while performing various remote jobs, by utilizing ground vehicles to carry and assist UAVs in various applications, including cargo delivery. This vehicle-drone cooperative cargo delivery mode benefits from both the large capacity of goods transportation of ground vehicles and the high mobility of UAVs.

In this paper, they employ multiple UAVs which are carried by one vehicle and manage to simultaneously deliver multiple parcels to customers residing in different locations. Figure 2.4 depicts a vehicle-drone cooperative cargo delivery scenario. There are roads near the nodes which refer to the customers. Candidate anchor points are at regular intervals on these roads. The anchor points are selected with the use of Facility Location Problem (FLP). UAVs are released from the vehicle at

each selected anchor point and visit customers for delivery along the planned paths, then return to the vehicle after the service. The delivery tasks for the nearby customers are abstracted as a Vehicle Routing Problem (VRP) and Traveling Salesman Problem (TSP) with a travel distance constraint in a single UAV path. The route distribution delivered to UAVs is formulated as a Bin Packing Problem (BPP). The purpose of this paper is to minimize the overall delivery time by path planning for UAVs and vehicles. This problem is a combinatorial optimization problem, and each sub-problem is NP-Hard.

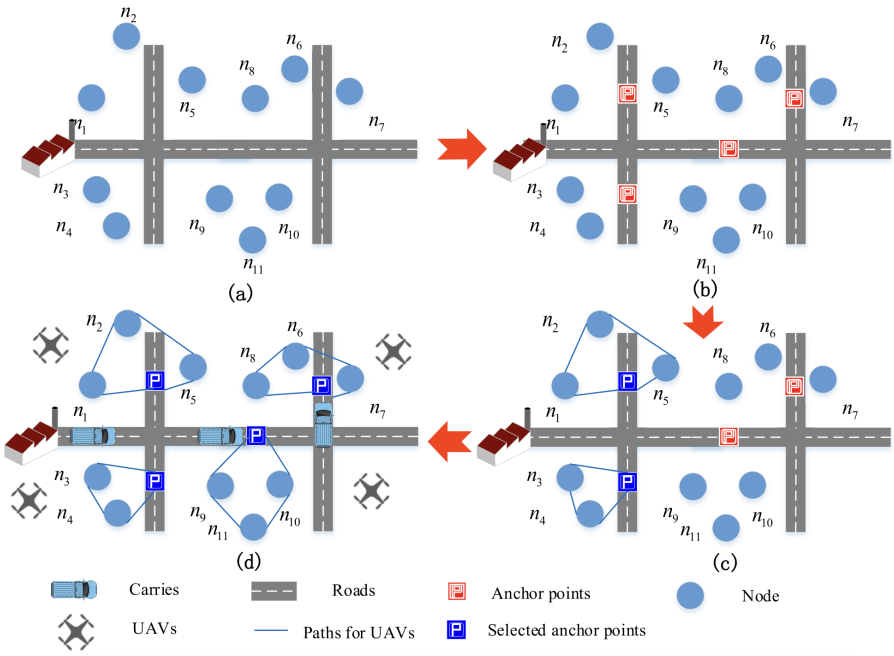

Figure 2.4: Illustration of the scenario where UAVs serve the customers with a carrier.

The paper proposes a novel Hybrid Genetic Algorithm (HGA). The algorithm consists of population management, population initialization, and population education. HGA can coordinate the complexity and the performance by jumping out the local optimum. Moreover, HGA applies to other problems with minor adjustments. In addition, they propose a Low Visit Cost Crossover algorithm (LVC) that selects genetic fragments based on nodes and anchor points, respectively. LVC ensures a better distribution in the population to avoid premature convergence. Furthermore, they also designed a three-hierarchical education algorithm, namely anchor point-based education, path-based education, and node-based education. Figure 2.5 depicts the flowchart of the algorithm.

The HGA initializes a population M which contains feasible solutions to the problem at first. Then, it iteratively selects two parent solutions P_1 and P_2 from M and generates a child solution with a smaller cost by the LVC Crossover Algorithm which is described in the later section. The local search algorithm is applied to

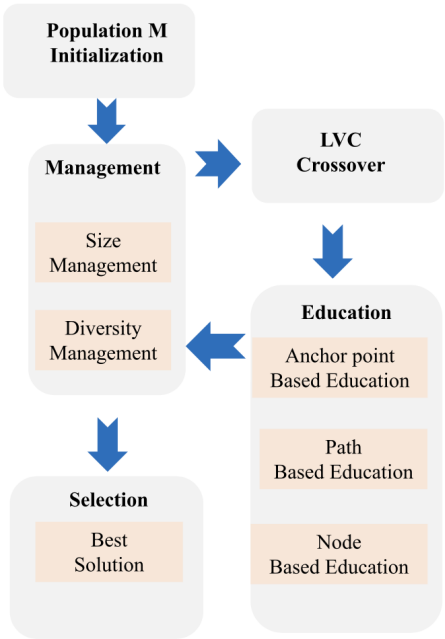

Figure 2.5: Flowchart of HGA.

optimize the child solution. M is updated when a better solution is obtained. To generate the population M, feasible solutions need to be constructed first. Optional anchor points are determined by the anchor point selection procedure. Then, the vehicle and UAV routes are obtained by a route construction procedure. Last, a scheduling strategy is adopted to reasonably allocate the UAV routes.

After the parent selection, the authors propose a novel Low Visit Cost Crossover (LVCC) algorithm to select the better gene fragment on the chromosome (solution) and compose a child solution with lower cost. The schematic diagram is shown in Figure 2.6. The crossover algorithm extracts all routes and all anchor points of P_1 and P_2 and stores them. The route selection procedure in LVCC iteratively selects appropriate routes from the pool of routes with the minimum cost. It also ensures that a customer shall not be included by multiple routes. The anchor point with the largest number of routes covered by it is selected. After the selection, we assign each route to the nearest selected anchor point to minimize the cost. The route for the vehicle to visit the selected anchor points is a TSP. The dominant gene of P_1 and P_2 is passed to the child C by the above steps.

After the LVCC algorithm, a better child solution C is obtained which inherits the dominant gene of P_1 and P_2. Then, the Local Search algorithm is adopted to replace the mutation factor of the traditional genetic algorithm and it includes three procedures: the anchor point-based education which deletes the anchor points which are not required, the route-based education merges multiple short routes to decrease distance and time costs, and the customer-based education, modifies the order in which the customers are served if it reduces distance and time costs.

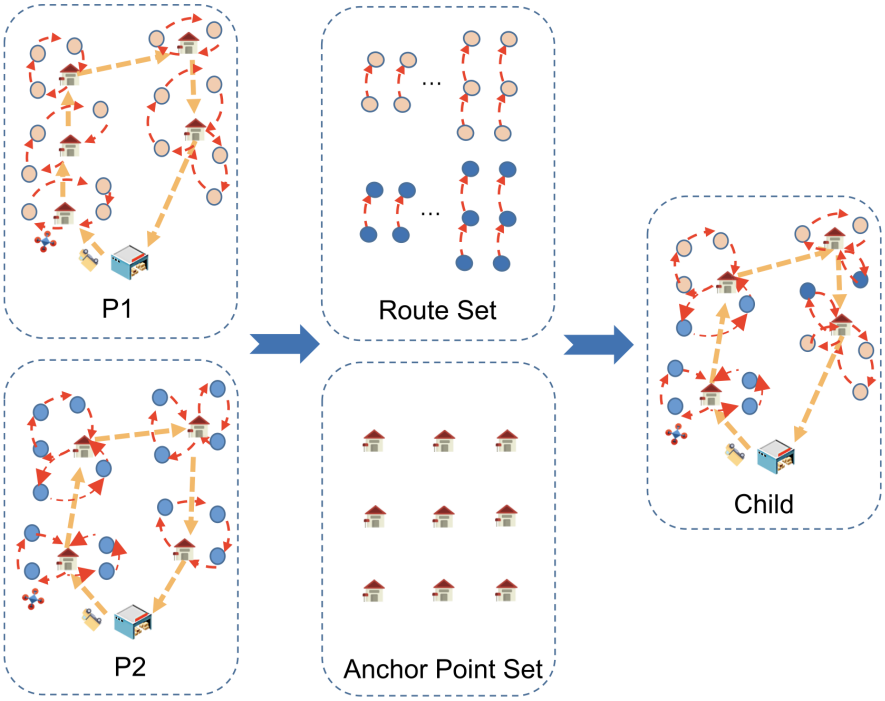

Figure 2.6: Illustration of LVC crossover algorithm.

To intuitively and overall analyze the performance and effect of the proposed algorithm, the authors utilize three algorithms as the baselines, namely the Lin-Kernighan Heuristic (LKH), a greedy algorithm (Greedy), and AMP (a relaxed version of AMP). HGA performed significantly better in all the settings and test cases. The major shortcoming of the algorithm is that it has not been tested in real-world simulations. Also, the number of drones to be carried is not specified, which is a problem when the number of customers is unknown.

The paper by Das et al. [4] introduces the expansion of Unmanned Aerial Vehicles (UAVs) from defense and security to commercial applications, particularly in the context of developing cost-efficient transport logistics solutions for the growing E-commerce and on-door services. The limitations of UAVs, such as their small size and weight capacity, are acknowledged, leading to the proposal of a novel mechanism. This mechanism involves integrating UAVs with delivery trucks, transforming the trucks into mobile launching and charging stations for UAVs while facilitating on-the-go parcel deliveries. The study recognizes the need for system-level research to optimize UAVs for last-mile delivery in the field of transportation logistics. Companies like Amazon and DHL are mentioned as pioneers in piloting parcel delivery systems with drone incorporation. The paper focuses on a synchronized delivery approach, formulating it as a multiple-objective mixed integer programming model. The primary objectives are to minimize travel distance and maximize customer satisfaction through scheduled deliveries within customer-defined time windows. To

address the unique challenges of the synchronized truck and drone routing problem, the paper introduces a Collaborative Pareto Ant Colony Algorithm (Collaborative P-ACO) and employs the Non-dominating Sort Genetic Algorithm II (NSGA-II) for evaluation.

The problem formulation section establishes the conditions and constraints for synchronized truck and drone routing in parcel delivery logistics. The goal is to ensure that each customer node is visited and delivered to exactly once, either by a UAV or a truck. Each delivery truck is assigned one UAV, working in coordination. The delivery path involves drone launching and retrieval at intermediate customer nodes along the truck's route, as shown in Figure 2.7. Time windows are predefined for each customer, and violation of these windows results in a loss of consumer satisfaction. Customers are categorized into groups (GR) based on their importance, and satisfaction loss due to time window violation is linked to the importance of the customer group.

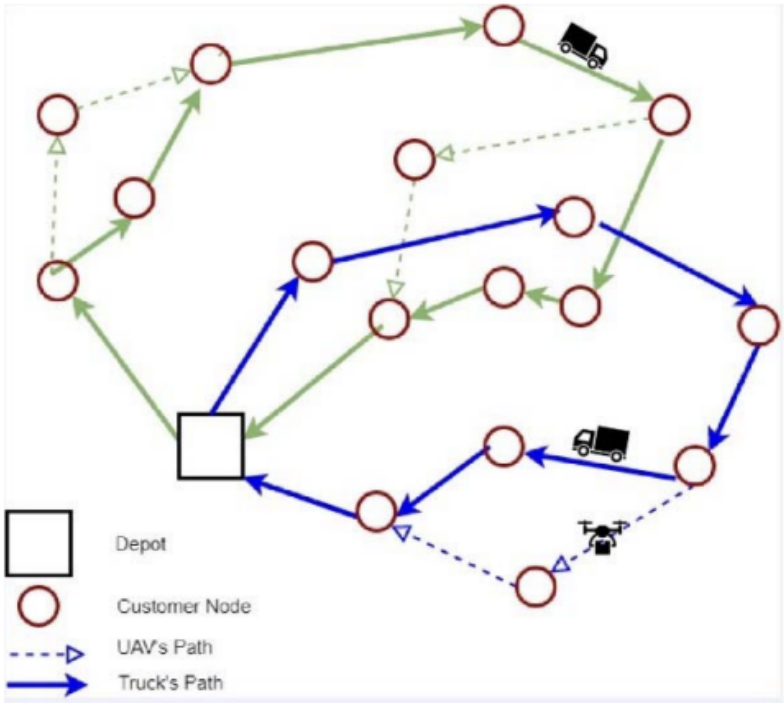

Figure 2.7: Vehicle routing problem with synchronized drones model.

Notational Convention: There are n customers with the starting and ending depot comprising a total of $n + 2$ nodes. The fleets of V_{size} trucks and D_{size} drones are represented by two sets, i.e., V and D. The n customers are divided into GR groups according to importance. Assumptions are as follows. Each customer is served only once by a truck or a drone. Drones can service only one customer node during a sub-tour, while trucks may visit multiple nodes during the drone sub-tour. Neither the truck nor the drone can stop at any point other than the nodes they are serving. Drones can be relaunched from the same node they were collected if

they were collected at a customer node. Drones are charged instantly after retrieval by the truck. The drone can only be retrieved at a customer node, not along an arc between two nodes in the truck's path. If a sub-tour ends at the depot, the drone is taken out of service. Different time window lengths are considered, each with a priority score. Multiple equally ranked time windows exist. There is one depot serving as the starting and ending point for the tour. The schedule covers the entire day's deliveries. Disruptions due to weather conditions or node capacity demands are known during routing. Trucks are in continuous uniform motion except at customer nodes, with no idle time on the arcs between customer nodes. Customers are divided into priority groups, ensuring better service levels for more important groups.

The methodology section of the paper discusses two algorithms, Collaborative P-ACO and NSGA II, to address the synchronized truck and drone routing problem. The Collaborative P-ACO is an enhancement of Pareto Ant Colony Optimization (ACO), designed for multi-objective problems. It introduces individual pheromones for each objective, employs random weights, and maintains an elite archive for solution selection. The Collaborative P-ACO differs from NSGA II in the process of solution generation. In P-ACO, a group of truck ants collaborates to find a feasible solution, starting from the depot. The truck paths are determined first, followed by the drone paths along the corresponding truck paths. The methodology outlines a two-stage formation of the ant path. In Stage 1, the formation of truck paths involves a group of truck ants moving simultaneously, selecting nodes based on pheromone levels and heuristic desirability. In Stage 2, drone paths are created along corresponding truck ant paths, considering endurance constraints. The paper introduces a novel parameter-tuning approach for P-ACO, adapting values from previous research and fine-tuning parameters through experimentation. The elite archive and scale control based on crowding distances are incorporated for better convergence rates. Additionally, the global pheromone update, evaporation, and parameter tuning details for P-ACO are provided. The values for heuristic desirables, initial pheromone levels, and other parameters are determined through trial and error. The methodology section concludes with the empirical determination of parameters like α, β, q_0, Q_c, Q_t, C, ρ, K, and $MaxIt$, balancing solution quality and computational time based on the complexity of the problem. The Non-Dominated Sort Genetic Algorithm-II (NSGA-II) proposed in this text focuses on optimizing a delivery system involving trucks and drones. The chromosome structure integrates information about trucks, drones, and customer nodes. The chromosome is divided into two parts, representing the customers served by trucks and drones, facilitating precise calculations of delivery and travel times. The algorithm considers route-feasibility constraints, ensuring sequential drone-truck order and adherence

to specific constraints during construction. The initialization involves selecting a random path for each truck, and the corresponding drone follows this path while serving nodes not covered by trucks. Trucks start from the depot, and depots are differentiated in representation to identify each truck's path. The algorithm utilizes non-domination sorting based on crowding distance, categorizing solutions into fronts and determining fitness within each front. Binary Tournament Selection is employed to choose parent chromosomes. Offspring generation involves crossover and mutation using the PMX operator and scramble operator, respectively. The chromosome structure is maintained during these operations, leading to possible offspring with fixed sizes based on customer nodes and vehicles. The NSGA-II demonstrates strong selectivity towards the fittest chromosomes, maintaining diversity with crowding distance. A controlled elitism mechanism is applied to ensure convergence to or near the global Pareto-optimal front, selecting a predefined number of individuals from different fronts based on their front and crowding distance.

The proposed mechanism of synchronized truck and drone delivery with launching and retrieval of drones on customer nodes on the truck path serves to be a very realistic and optimal solution in the scenario of delivery logistics accounting for both cost minimization and maximization of in-time deliveries. The insights from the results suggest that the use of drones for parcel delivery serves to be a very efficient method in terms of travel costs and customer service. The proposed Collaborative P-ACO algorithm performs well on different instances as compared to the NSGA-II and proves to be a computationally inexpensive meta-heuristic for the truck-drone routing problem. The feasibility of the use of drones in real-life scenarios for use in populated regions faces some issues of security and safety. The model is limited to deterministic demand and fixed truck and drone pairs. More flexibility in terms of a nonuniform fleet of trucks and drones, stochastic changes in demand and launch and retrieval at any points along the arc between two nodes on the truck's path and exchange of drones between trucks also need to be included.

Chapter 3

Methodology

3.1 Problem Statement

3.1.1 Mathematical Formulation

For every geographic region (say, per post office), we have:

- 1 warehouse, W, with known $x - y$ coordinates.
- n drone pickup points (d_1 through d_n), with known $x - y$ coordinates.

For every drone pickup point d_i, we have:

- A variable number of customer residences surrounding d_i, with known $x - y$ coordinates.
- 1 drone servicing each drone pickup point d_i, that sits at the pickup point when idle.

We also have a set of participating third-party cabs whose routes are known to us in advance.

Our aim is to deliver the parcels from the warehouse W to the end customers residing in the service-zone of at least one drone pickup point, by having third-party cabs deliver the selected parcel(s) from the warehouse to the selected pickup point, and then have the drone at that pickup point perform the final delivery from the pickup point to the end customer.

3.1.2 Assumptions

1. Packages are of unit volume and mass in a hypothetical unit.
2. All cabs have infinite driving range (within the considered geographic region).

3. The warehouse has infinite capacity for housing parcels.
4. All drones can move unrestricted through the Euclidean space of their flying range.
5. All considered cabs go past the central warehouse of the considered region, and at least one pickup point.

3.1.3 Constraints

- Known route of each cab.
- Capacity of each drone pickup point.
- Capacity of cab.
- Capacity of drones.
- Maximum flying range of each drone on a full battery.

3.1.4 Decision Parameters

- Given a cab carrying a parcel(s), which drone pickup point(s) should that parcel(s) be assigned to for drop off by the cab?
- Given a parcel request from a customer residence, which cab should that parcel be assigned to for picking up from the warehouse?
- Given a set of currently dropped off parcels at a pickup point, what order of deliveries to follow using the drone?
- Given a drone flying with a set of parcels, in what order should the respective customer residences be visited before returning to the pickup point?

3.1.5 Optimization Problem

We mathematically formulate our problem as a minimization of the objective function in Equation 1, subject to constraints shown in Equations 2 - 4 below:

minimize:

$$\sum_{\forall i} distance\,(cab_i) + \sum_{\forall j} distance\,(drone_j) \tag{1}$$

such that:

$$distance\,(drone_j) \leq range\,(drone_j)\,\forall j \tag{2}$$

$$load\,(drone_j) \leq capacity\,(drone_j)\,\forall j \tag{2}$$

$$load\,(cab_i) \leq capacity\,(cab_i)\,\forall i \tag{3}$$

$$load\,(pickup_k) \leq capacity\,(pickup_k)\,\forall k \tag{4}$$

3.2 Proposed Solution

We propose the following two algorithms as a solution to the proposed problem – Algorithm 1 to conduct the delivery of parcels from the warehouse to the selected pickup point and Algorithm 2 to conduct the delivery of parcels at a particular pickup point to the end customers.

Algorithm 1 Greedily assign cabs for parcels requested by customers.

```
procedure SelectCab(W_x, W_y)
    r ← a mod b
    for all r_i do
        Select the closest pickup point p_k, such that residence r_i is in the serviceable zone of p_k.
        Select the cab cab_j that passes by p_k the earliest of all known cab routes; if such a cab does not exist, continue for next r_i.
        Assign requesting residence r_i's parcel parcel_i to selected cab cab_j if its capacity constraint is satisfied, parcel_i → cab_j.
        If capacity of cab is exceeded, assign parcel to next earliest cab, and so on.
    end for
    return {parcel_i → cab_j}
end procedure
```

Algorithm 1 takes as input:

- Coordinates of warehouse, (W_x, W_y).
- Coordinates of all pickup points, $(p_{k,x}, p_{k,y})\,\forall k$.
- Set of parcel requests currently unfulfilled, $\{parcel_i\}$, with their corresponding coordinates of requesting residences, $(r_{i,x}, r_{i,y})$.
- Set of cabs currently known to pass by W and at least one p_k, $\{cabs_j\}$.

and produces as output:

- A set of matchings from parcels to cabs, $\{parcel_i \rightarrow cab_j\}$ (not necessarily for all i, i.e., some parcels may remain unmatched).

Algorithm 2 Define the route to be taken by drone and then perform TSP on selected parcels

1: **procedure** SELECTDRONEROUTE(p_k)
2: Select the earliest dropped c number of parcels (intended for residences), where c is the capacity of the drone at p_k
3: Select $parcel_i$, say $parcel_{farthest}$ which is farthest from p_k ($\sqrt{(r_{i,x} - p_{k,x})^2 + (r_{i,y} - p_{k,y})^2}$ where $parcel_i$ is intended for r_i)
4: $radius \leftarrow \sqrt{(r_{farthest,x} - p_{k,x})^2 + (r_{farthest,y} - p_{k,y})^2}$
5: Create $sector_\theta$ with angle θ such that $radius \times \theta + 2 \times radius = range_k$ where $range_k$ is the flying range of drone stationed at p_k
6: Create $[r_i]$ such that r_i inside $sector_\theta$
7: Create a fully connected weighted graph with $(p_{k,x}, p_{k,y})$ and $(r_{i,x}, r_{i,y})$ as nodes with weights being the Euclidean distance between two nodes
8: Perform a Christofides Approximation of TSP on the constructed graph, specifying the start and end node to necessarily be $(p_{k,x}, p_{k,y})$
9: **return** TSP route determined
10: **end procedure**

Algorithm 2 takes as input:

- Coordinates of considered pickup point, $(p_{k,x}, p_{k,y})$.
- Set of parcels currently dropped off at the considered pickup point with their corresponding coordinates of requesting residences, $(r_{i,x}, r_{i,y})$

and produces as output

- An ordered list of residences $[r_i]$ for the drone to visit for delivery in each cycle (before returning to pickup point for next cycle).

In this algorithm, we use Christofides [5] approximation of the Travelling Salesman Problem (TSP), which provides us with a route that is guaranteed to be within $\frac{3}{2}$ of the total length of the optimal TSP tour.

The fact that our sector construction helps in determining a TSP tour which does not exceed the flying range of the drone is highlighted via Figure 3.1 and Figure 3.2, which show the same residences being serviced by a drone at the same pickup point in both cases. In both cases, the flying range of the drone is considered to be 300 units.

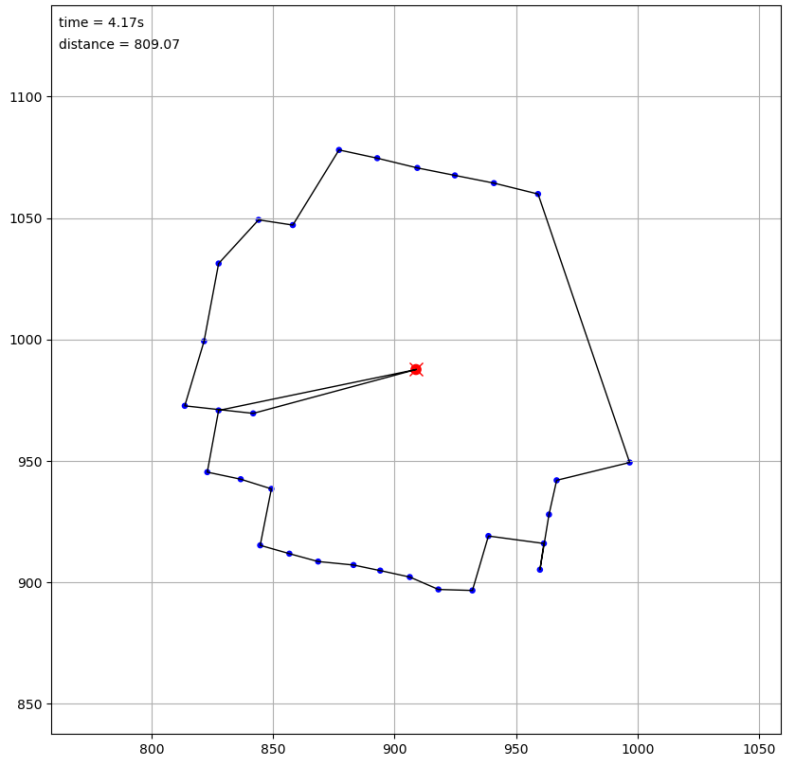

Figure 3.1: TSP tour without sector construction, which exceeds the flying range of the drone (300 units).

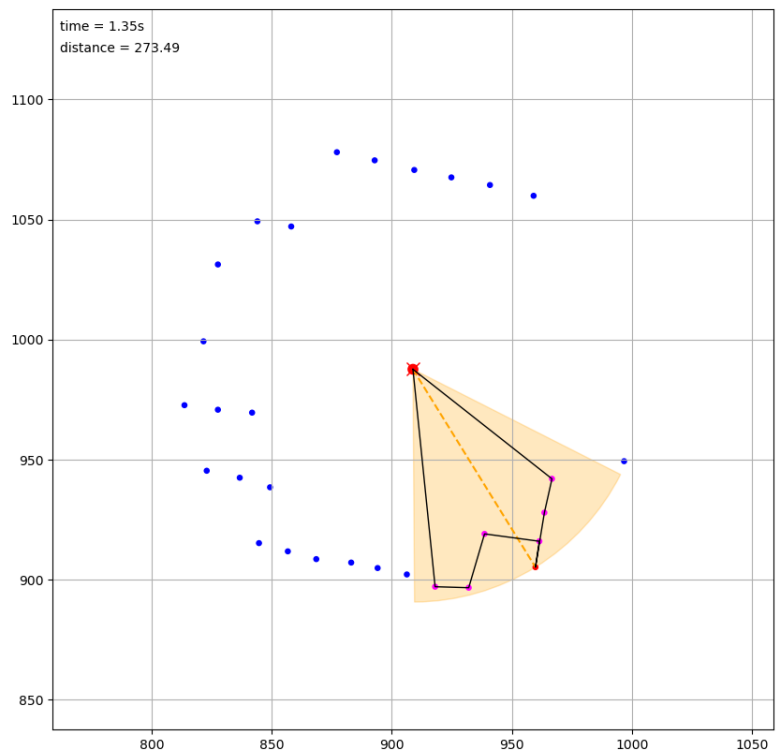

Figure 3.2: TSP tour with sector construction, which does not exceed the flying range of the drone (300 units).

Chapter 4

Experiments

4.1 Experimental Setup

4.1.1 Tool Setup

We use the tool SUMO [6] for all simulations of vehicles and traffic conditions. We also use TraCI[7] along with Python 3.11.5 to interface with SUMO and use our custom logic and algorithms in the simulations.

Figure 4.1: Map instance considered for testing.

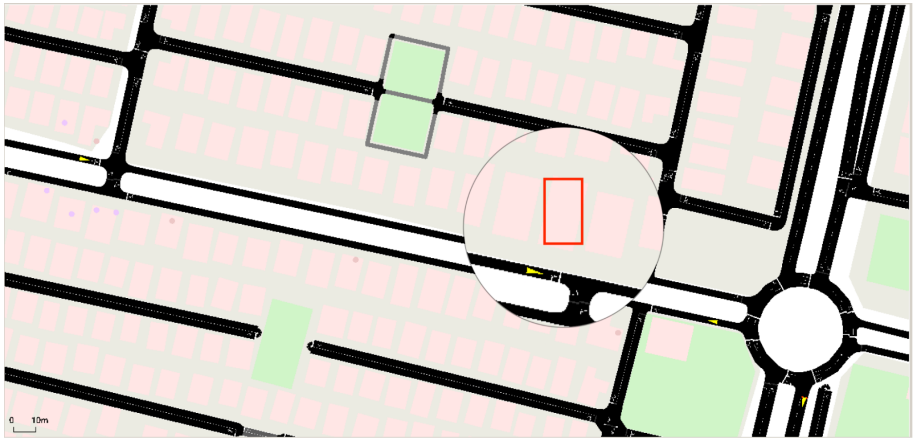

Figure 4.2: Warehouse highlighted in zoomed-in image of map instance.

4.1.2 Data Used

We use map data from Open Street Maps [8] for use in simulations with SUMO, as depicted in Figure 4.1, a zoomed version of which is shown in Figure 4.2.

Figure 4.3 shows another zoomed instance of a running simulation, where a drone can be seen currently in-flight. The red region denotes the pickup point for that drone, while the blue region denotes the warehouse building. Purple buildings

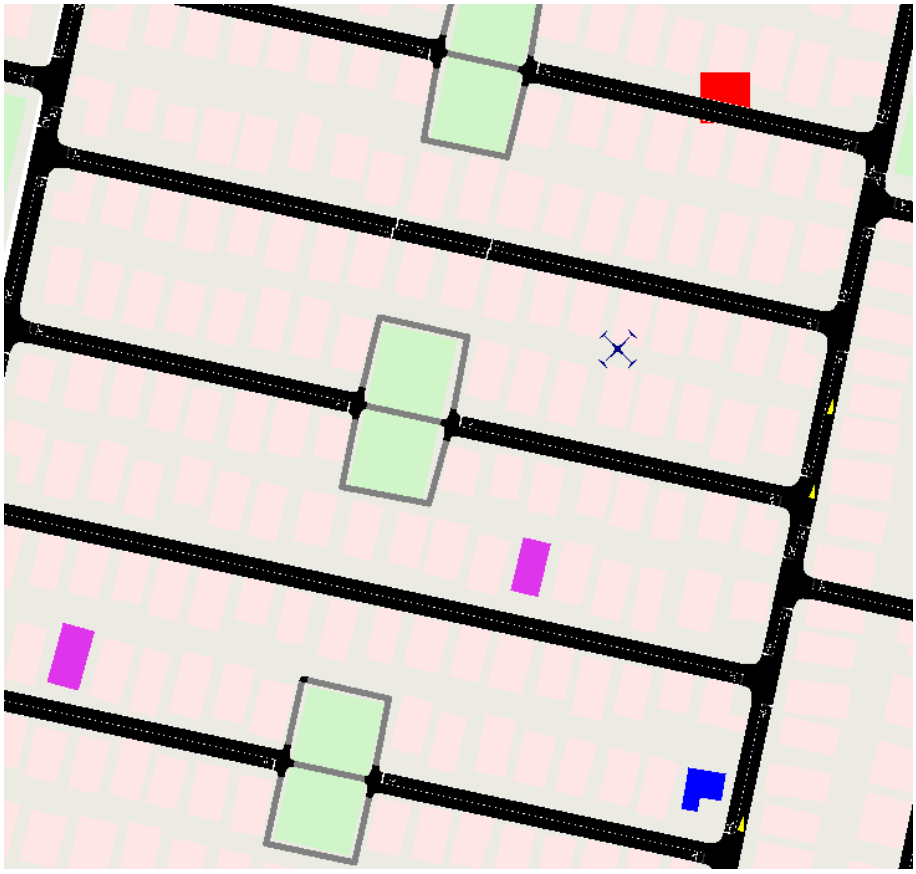

Figure 4.3: A still capture from a running simulation.

are the residences from where parcel requests have arrived from. Along the roads closest to the warehouse, several cabs can be seen which have been denoted as yellow triangles.

4.2 Simulation Code

Python code written by us for conducting the experimental simulations is available at [1], under the BSD-3-Clause license.

Additionally, all documentation pertaining to the Application Programming Interface (API) of our Python module(s) is available at [2].

Instructions for running our code follow: Visit [3] and follow the instructions there to install SUMO.

1. Clone our repository:
   `$ git clone "https://github.com/pganguli/drone-cab.git"`

2. After changing directory into the cloned repository, run:
   `$ poetry install --extras docs`
   to install all necessary Python dependencies.

3. Run the main simulation script with:
   `$ poetry run python main.py`

[1] https://github.com/pganguli/drone-cab
[2] https://drone-cab.netlify.app/py-modindex
[3] https://sumo.dlr.de/docs/Installing/index.html

Chapter 5

Conclusion

Delivery of parcels in a smart city certainly can be improved by the implementation and coordination of a hybrid network of vehicles such as the one proposed by our work. This network however also needs the support of the existing infrastructure, or new infrastructure has to be built in order to enable the deployment of the networking aspects of such systems.

The placement of a minimal number of pickup points for the drones to be able to service all buildings in a geographic region also remains to be optimized, which can be explored in some future work. The simplifying assumptions of having no airborne obstacles for the drones also needs to be eliminated in a real-world scenario, and traffic conditions of the road network needs to be taken under consideration when designating third-party cabs to specific parcels.

In order to prevent the problem of starvation, where some parcels may never end up being delivered due to the unavailability of third-party cabs going in the direction of the residence where the request came from, we propose the use of a small pool of first-party owned cabs or trucks which can perform such deliveries once enough such parcels have accumulated. Such a delivery will be scheduled and carried out as regular deliveries by the postal service is currently performed.

We also note that our proposed algorithm can be extended to accommodate multiple drones to a single pickup point for servicing that region. The scheduling and assignment of drones and cabs in such a scenario will have to be formulated as a multi-objective optimization problem, and the implementation has to deal with the parallel scheduling of these drones per pickup point.

Overall, we anticipate a great number future works to adopt strategies similar to the ones we have outlined, and build upon the data-caching inspired network of vehicles proposed in our given solution to the problem of parcel delivery in smart cities.

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